

TEST DEMOGRAPHIC & SERVICE ASSOCIATION WITH CHURN USING PEARSON CHI-SQUARE TEST OF INDEPENDENCE

RESULT

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Contract * churn_num	7043	100.0%	0	0.0%	7043	100.0%

Contract * churn_num Crosstabulation

			churn_num		Total
			0	1	
Contract	Month-to-month	Count	2220	1655	3875
		% within Contract	57.3%	42.7%	100.0%
		Adjusted Residual	-34.0	34.0	
	One year	Count	1307	166	1473
		% within Contract	88.7%	11.3%	100.0%
		Adjusted Residual	14.9	-14.9	
	Two year	Count	1647	48	1695
		% within Contract	97.2%	2.8%	100.0%
		Adjusted Residual	25.4	-25.4	
	Total	Count	5174	1869	7043
		% within Contract	73.5%	26.5%	100.0%

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	1184.597 ^a	2	.000
Likelihood Ratio	1386.810	2	.000
N of Valid Cases	7043		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 390.89.

Symmetric Measures

	Value	Approx. Sig.
Nominal by Nominal Phi	.410	.000
Cramer's V	.410	.000
N of Valid Cases	7043	

a. Not assuming the null hypothesis.

b. Using the asymptotic standard error assuming the null hypothesis.

INTERPRETATION

A Pearson Chi-Square test of independence was performed to examine the relation between contract type (Contract) and customer churn (churn_num). The relation between these variables was statistically significant, $\chi^2(2, N = 7043) = 1184.60$, $p < .001$, with a moderate-to-strong effect size (Cramér's $V = .410$).

COMPARE SPENDING ACROSS CHURN STATUS USING INDEPENDENT SAMPLE t-TEST

RESULT

Group Statistics

	churn_num	N	Mean	Std. Deviation	Std. Error Mean
MonthlyCharges	0	5174	61.2651	31.09265	.43226
	1	1869	74.4413	24.66605	.57055

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
MonthlyCharges	Equal variances assumed	358.128	.000	-16.537	7041	.000	-13.17621	.79678	-14.73815	-11.61427
	Equal variances not assumed			-18.408	4135.795	.000	-13.17621	.71581	-14.57957	-11.77284

INTERPRETATION

An independent samples *t*-test was conducted to compare monthly charges between customers who churned and those who remained. Levene's test indicated unequal variances ($F = 358.13, p < .001$), so degrees of freedom were adjusted via Welch's correction

The results indicated a statistically significant difference in monthly charges, $t(4135.80) = -18.41, p < .001$, Cohen's $d = 0.45$. Customers who churned had significantly higher monthly charges ($M = \$74.44, SD = 24.67$) than retained customers ($M = \$61.27, SD = 31.09$), representing a mean premium of \$13.18, 95% CI [$-\$14.58, -\11.77].

TIME-TO-CHURN SURVIVAL ANALYSIS USING KAPLAN-MEIER SURVIVAL CURVE & COX PROPORTIONAL HAZARDS.

RESULT

Means and Medians for Survival Time

Contract	Mean ^a				Median			
	Estimate	Std. Error	95% Confidence Interval		Estimate	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound			Lower Bound	Upper Bound
Month-to	18.037	.284	17.480	18.594	12.000	.388	11.240	12.760
One year	42.045	.496	41.073	43.017	44.000	.855	42.323	45.677
Two year	56.735	.442	55.868	57.602	64.000	.517	62.986	65.014
Overall	32.371	.293	31.798	32.945	29.000	.586	27.852	30.148

a. Estimation is limited to the largest survival time if it is censored.

Overall Comparisons

	Chi-Square	df	Sig.
Log Rank (Mantel-Cox)	3774.212	2	.000
Breslow (Generalized Wilcoxon)	3280.796	2	.000
Tarone-Ware	3592.287	2	.000

Test of equality of survival distributions for the different levels of Contract.

INTERPRETATION

A Kaplan-Meier survival analysis was conducted to compare customer retention distributions across contract types (Contract). A Log-Rank (Mantel-Cox) test indicated a statistically significant difference in survival profiles between groups, $\chi^2(2) = 3774.21, p < .001$. Month-to-month subscribers exhibited a median survival time of 12.00 months (95% CI [11.24,12.76]), whereas customers on one-year and two-year contracts exhibited median survival times of 44.00 months (95% CI [42.32,45.68]) and 64.00 months (95% CI [62.99,65.01]), respectively.

PREDICTIVE MODELING USING BINARY LOGISTIC REGRESSION

RESULT

Case Processing Summary

Unweighted Cases ^a		N	Percent
Selected Cases	Included in Analysis	7043	100.0
	Missing Cases	0	.0
	Total	7043	100.0
Unselected Cases		0	.0
Total		7043	100.0

a. If weight is in effect, see classification table for the total number of cases.

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	2080.323	5	.000
	Block	2080.323	5	.000
	Model	2080.323	5	.000

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	6069.822 ^a	.256	.373

a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

Classification Table^a

		Predicted		Percentage Correct
		churn_num 0	1	
Step 1	Observed churn_num 0	4647	527	89.8
	1	953	916	49.0
Overall Percentage				79.0

a. The cutvalue is .500

Variables in the Equation

	B	S.E.	Wald	df	Sig.	Exp(B)	
Step 1 ^a	tenure	-.037	.002	325.181	1	.000	.963
	MonthlyCharges	.025	.001	318.460	1	.000	1.025
	contract_num	-.934	.072	166.428	1	.000	.393
	SeniorCitizen	.431	.081	28.555	1	.000	1.538
	PaperlessBilling(1)	-.467	.072	42.522	1	.000	.627
	Constant	-.349	.126	7.625	1	.006	.705

a. Variable(s) entered on step 1: tenure, MonthlyCharges, contract_num, SeniorCitizen, PaperlessBilling.

INTERPRETATION

A binary logistic regression was performed to ascertain the effects of customer tenure, monthly charges, contract type, senior citizen status, and paperless billing on the likelihood of churn. The logistic regression model was statistically significant, $\chi^2(5) = 2080.323$, $p < .001$. The model explained 37.3% (Nagelkerke R^2) of the variance in churn and correctly classified 79.0% of cases.

Increasing contract duration ($\text{Exp}(B) = 0.393$, $p < .001$) and customer tenure ($\text{Exp}(B) = 0.963$, $p < .001$) significantly decreased the odds of churn. Conversely, higher monthly charges ($\text{Exp}(B) = 1.025$, $p < .001$) and senior citizen status ($\text{Exp}(B) = 1.538$, $p < .001$) significantly elevated churn probability.

Z-TEST A/B COHORT & FEATURE ADOPTION ANALYSIS

RESULT

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Numeric * TechSupport	7043	100.0%	0	0.0%	7043	100.0%

Numeric * TechSupport Crosstabulation

			TechSupport			Total
			No	No internet service	Yes	
Numeric	1.00	Count	2680 ^a	524 ^b	671 ^b	3875
		% within TechSupport	77.2%	34.3%	32.8%	55.0%
	2.00	Count	557 ^a	364 ^b	552 ^c	1473
		% within TechSupport	16.0%	23.9%	27.0%	20.9%
	3.00	Count	236 ^a	638 ^b	821 ^b	1695
		% within TechSupport	6.8%	41.8%	40.2%	24.1%
Total	Count	3473	1526	2044	7043	
	% within TechSupport	100.0%	100.0%	100.0%	100.0%	

Each subscript letter denotes a subset of TechSupport categories whose column proportions do not differ significantly from each other at the .05 level.

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	1543.471 ^a	4	.000
Likelihood Ratio	1657.204	4	.000
N of Valid Cases	7043		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 319.15.

INTERPRETATION

A Pearson Chi-Square test of independence with Bonferroni-adjusted pairwise column proportion comparisons revealed a statistically significant association between contract commitment level and technical support adoption, $\chi^2(4, N = 7043) = 1543.47, p < .001$, with a moderate effect size (Cramér's $V = .331$). The analysis demonstrated a pronounced service gap in short-term accounts, where month-to-month subscribers exhibited a 77.2% ($n = 2,680$) rate of not having technical support and only a 32.8% ($n = 671$) adoption rate—proportions that were significantly lower than those observed among one-year (27.0% support adoption, $n = 552$) and two-year contract holders (40.2% support adoption, $n = 821$)—confirming that customer attrition risk is heavily compounded by the absence of value-added onboarding services in flexible billing tiers.